EXPERIMENT - 67

1. Problem Statement

Write a recursive function to find the **GCD of two positive integers using Euclid's algorithm**.

2. Algorithm

Main Program

Step 1: START
Step 2: Read the two positive integers a and b.
Step 3: Call gcd(a, b) and display the result.
Step 4: STOP

Function gcd(a, b)

Step 1: If b = 0, return a.
Step 2: Return gcd(b, a % b).

3. Test Case Table

#	First Number	Second Number	Expected Output	Actual Output	Result
1	48	18	GCD: 6	GCD: 6	**PASS**
2	17	5	GCD: 1	GCD: 1	**PASS**
3	100	25	GCD: 25	GCD: 25	**PASS**
4	7	7	GCD: 7	GCD: 7	**PASS**
5	12	36	GCD: 12	GCD: 12	**PASS**

4. Source Code

gcd67.c

```
/* Experiment 67: GCD of two numbers using Euclid's algorithm and recursion. */
#include <stdio.h>

int gcd(int a, int b);

int main(void)
{
    int a, b;

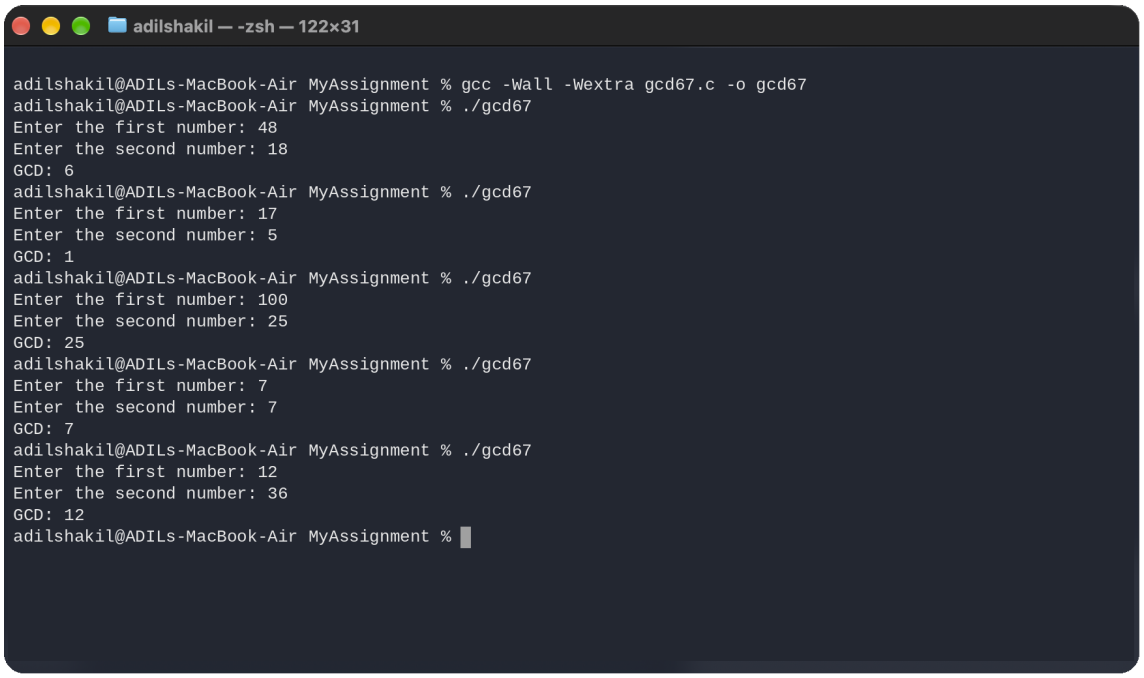
    printf("Enter the first number: ");
    scanf("%d", &a);
    printf("Enter the second number: ");
    scanf("%d", &b);

    printf("GCD: %d\n", gcd(a, b));
    return 0;
}

/* gcd(a, b) is the same as gcd(b, a % b). When b becomes 0,
   a is the answer. */
int gcd(int a, int b)
{
    if (b == 0)
        return a;
    return gcd(b, a % b);
}
```

5. Compilation and Output

Compiled with `gcc -Wall -Wextra gcd67.c -o gcd67` so the small warnings would show up too. It built clean.



All five passed.